



Hands-On Activity: Data analysis and formulas: Bakery sales insights

Activity Overview

By now, you've been introduced to the basics of using formulas to perform calculations for data analysis. In this activity, you will edit the Bakery Sales spreadsheet you created earlier in the [Hands-On activity: Introduction to Google Sheets](#). You will revisit the data and review the new data, then use formulas to calculate key metrics and gain insights. By the time you complete this activity, you will be more familiar with using simple formulas to analyze and extract meaningful information from datasets. Knowing how to use formulas in spreadsheet applications is an essential skill for any data analyst, as they help you automate calculations, make data-driven decisions, and save time.

Step-By-Step Instructions

Follow the instructions to complete each step of the activity. Then answer the question at the end of the activity before going to the next course item.

Step 1: Access the spreadsheet

To get started, determine which software you'd like to use, such as Google Sheets or Microsoft Excel. You will also need the updated [Bakery Sales March 2020](#) spreadsheet, which has new data that was not in the previous activity.

To use the template for this course item, click the link below and select “Use Template.”

Template: [Bakery Sales March 2020](#)

OR

If you don't have a Google account, download the template directly from the attachment below.

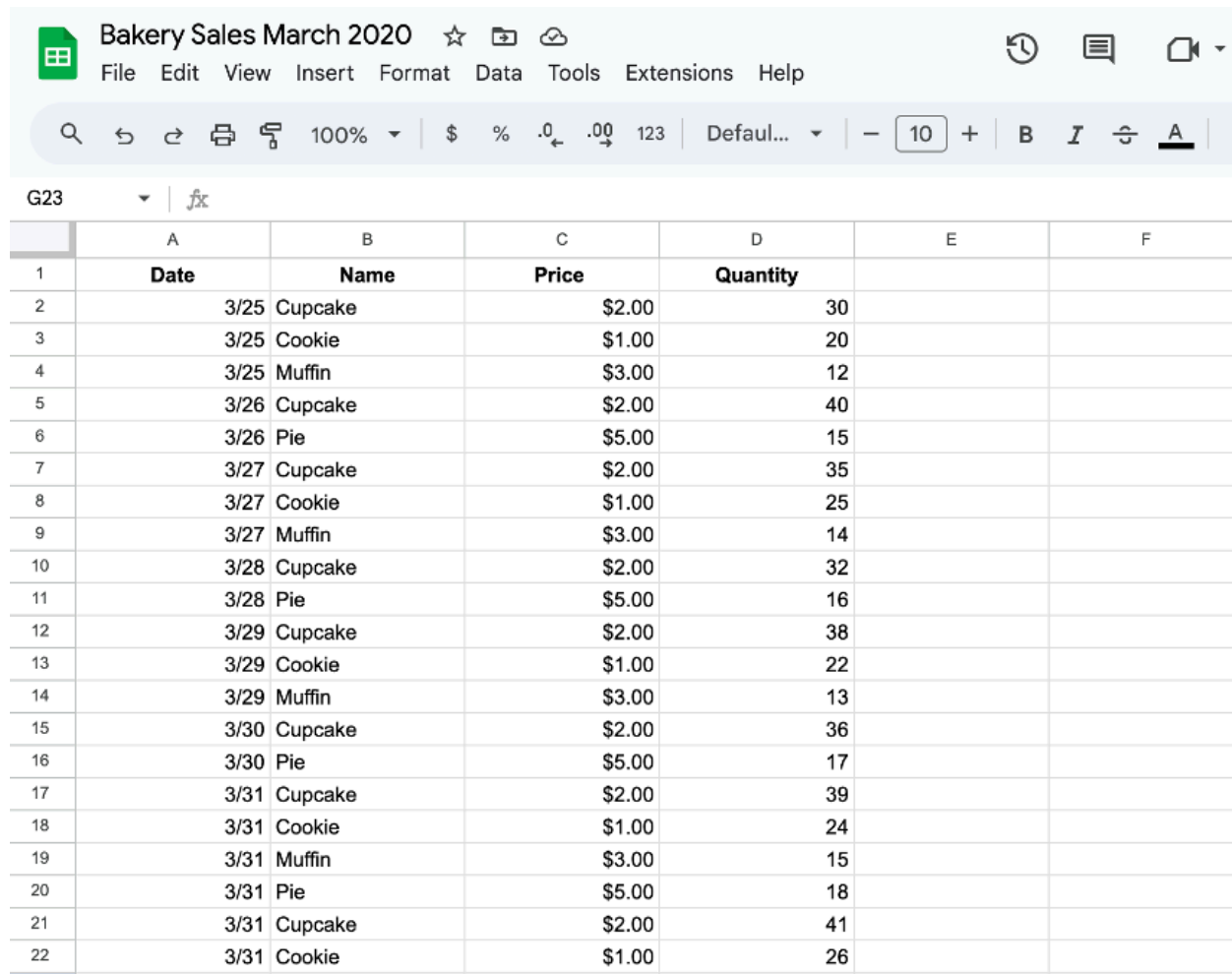
[Bakery Sales March 2020](#)

[XLSX File](#)

Step 2: Edit an existing preadsheet

The new Bakery Sales March 2020 spreadsheet has been updated with the sales data from the local bakery for the remaining days of the month. In this activity, you'll use formulas to calculate essential metrics that reveal the revenue generated by each

product. This data analysis will provide valuable insights into the local bakery's performance.



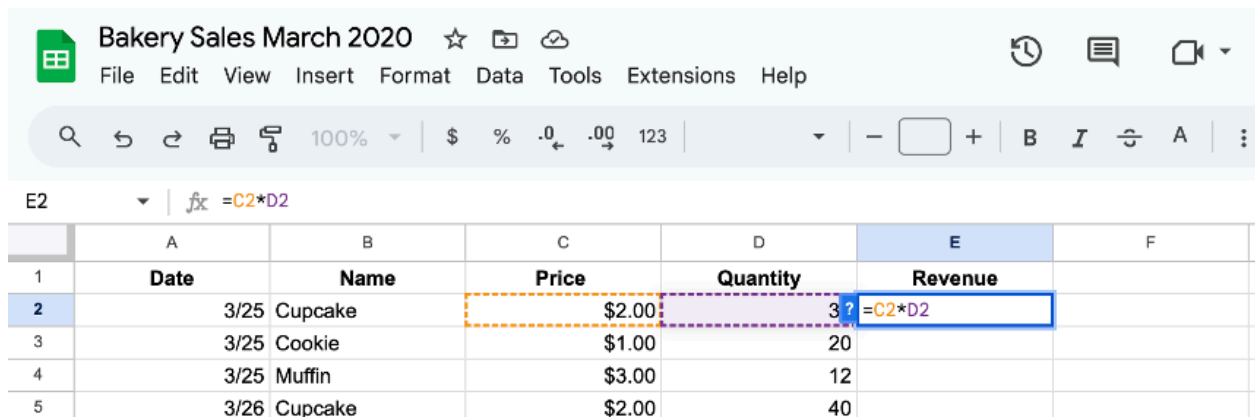
The screenshot shows a Google Sheets interface with a spreadsheet titled "Bakery Sales March 2020". The spreadsheet contains a table with 22 rows of data. The columns are labeled A through F. Column A is "Date", Column B is "Name", Column C is "Price", and Column D is "Quantity". Columns E and F are empty. The data represents daily sales for four products: Cupcake, Cookie, Muffin, and Pie, from March 25 to March 31, 2020.

	A	B	C	D	E	F
1	Date	Name	Price	Quantity		
2	3/25	Cupcake	\$2.00	30		
3	3/25	Cookie	\$1.00	20		
4	3/25	Muffin	\$3.00	12		
5	3/26	Cupcake	\$2.00	40		
6	3/26	Pie	\$5.00	15		
7	3/27	Cupcake	\$2.00	35		
8	3/27	Cookie	\$1.00	25		
9	3/27	Muffin	\$3.00	14		
10	3/28	Cupcake	\$2.00	32		
11	3/28	Pie	\$5.00	16		
12	3/29	Cupcake	\$2.00	38		
13	3/29	Cookie	\$1.00	22		
14	3/29	Muffin	\$3.00	13		
15	3/30	Cupcake	\$2.00	36		
16	3/30	Pie	\$5.00	17		
17	3/31	Cupcake	\$2.00	39		
18	3/31	Cookie	\$1.00	24		
19	3/31	Muffin	\$3.00	15		
20	3/31	Pie	\$5.00	18		
21	3/31	Cupcake	\$2.00	41		
22	3/31	Cookie	\$1.00	26		

The local bakery manager wants to know how many of each product was sold and how much revenue each product brought in during this sales period. To find the answers, follow these steps:

1. Create a new attribute named **Revenue** in cell **E1**. Bold and center this column name like you did previously.

2. To calculate the revenue for the 30 cupcakes sold on 3/25, select cell **E2**. Next, multiply the number of cupcakes sold by the price of each cupcake. Enter **=C2*D2** and press **Enter**.



The screenshot shows a Google Sheets document titled "Bakery Sales March 2020". The spreadsheet has the following data:

	A	B	C	D	E	F
	Date	Name	Price	Quantity	Revenue	
2	3/25	Cupcake	\$2.00	30	=C2*D2	
3	3/25	Cookie	\$1.00	20		
4	3/25	Muffin	\$3.00	12		
5	3/26	Cupcake	\$2.00	40		

The formula bar at the top shows the formula **=C2*D2** for cell E2. The cell E2 is highlighted with a blue border.

3. Copy the formula (or use the fill handle) from cell **E2** to **E3:E22**.

Bakery Sales March 2020

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	A	B	C	D	E	F
1	Date	Name	Price	Quantity	Revenue	
2	3/25	Cupcake	\$2.00	30	\$60.00	
3	3/25	Cookie	\$1.00	20	\$20.00	
4	3/25	Muffin	\$3.00	12	\$36.00	
5	3/26	Cupcake	\$2.00	40	\$80.00	
6	3/26	Pie	\$5.00	15	\$75.00	
7	3/27	Cupcake	\$2.00	35	\$70.00	
8	3/27	Cookie	\$1.00	25	\$25.00	
9	3/27	Muffin	\$3.00	14	\$42.00	
10	3/28	Cupcake	\$2.00	32	\$64.00	
11	3/28	Pie	\$5.00	16	\$80.00	
12	3/29	Cupcake	\$2.00	38	\$76.00	
13	3/29	Cookie	\$1.00	22	\$22.00	
14	3/29	Muffin	\$3.00	13	\$39.00	
15	3/30	Cupcake	\$2.00	36	\$72.00	
16	3/30	Pie	\$5.00	17	\$85.00	
17	3/31	Cupcake	\$2.00	39	\$78.00	
18	3/31	Cookie	\$1.00	24	\$24.00	
19	3/31	Muffin	\$3.00	15	\$45.00	
20	3/31	Pie	\$5.00	18	\$90.00	
21	3/31	Cupcake	\$2.00	41	\$82.00	
22	3/31	Cookie	\$1.00	26	\$26.00	

4. To find the total number of each product sold, first calculate the total quantity of each one. Start by making the data clear and easy to read. In cell **G3**, enter **Total # of Cookies Sold**. In cell **G4**, enter **Total # of Cupcakes Sold**. In cell **G5**, enter **Total # of Muffins Sold**. In cell **G6**, enter **Total # of Pies Sold**.

5. Now, calculate the total quantity of cookies sold. Select cell **H3**, enter **=D3+D8+D13+D18+D22** and press **Enter**.

Bakery Sales March 2020

	A	B	C	D	E	F	G	H
	Date	Name	Price	Quantity	Revenue			
2	3/25	Cupcake	\$2.00	30	\$60.00			
3	3/25	Cookie	\$1.00	20	\$20.00		Total # of Cookies Sold	=D3+D8+D13+D18+D22
4	3/25	Muffin	\$3.00	12	\$36.00		Total # of Cupcakes Sold	
5	3/26	Cupcake	\$2.00	40	\$80.00		Total # of Muffins Sold	
6	3/26	Pie	\$5.00	15	\$75.00		Total # of Pies Sold	
7	3/27	Cupcake	\$2.00	35	\$70.00			
8	3/27	Cookie	\$1.00	25	\$25.00			
9	3/27	Muffin	\$3.00	14	\$42.00			
10	3/28	Cupcake	\$2.00	32	\$64.00			
11	3/28	Pie	\$5.00	16	\$80.00			
12	3/29	Cupcake	\$2.00	38	\$76.00			
13	3/29	Cookie	\$1.00	22	\$22.00			
14	3/29	Muffin	\$3.00	13	\$39.00			
15	3/30	Cupcake	\$2.00	36	\$72.00			
16	3/30	Pie	\$5.00	17	\$85.00			
17	3/31	Cupcake	\$2.00	39	\$78.00			
18	3/31	Cookie	\$1.00	24	\$24.00			
19	3/31	Muffin	\$3.00	15	\$45.00			
20	3/31	Pie	\$5.00	18	\$90.00			
21	3/31	Cupcake	\$2.00	41	\$82.00			
22	3/31	Cookie	\$1.00	26	\$26.00			
23								

6. Then, calculate the total quantity of cupcakes, muffins, and pies sold by using the same steps.

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	A	B	C	D	E	F	G	H
1	Date	Name	Price	Quantity	Revenue			
2	3/25	Cupcake	\$2.00	30	\$60.00			
3	3/25	Cookie	\$1.00	20	\$20.00		Total # of Cookies Sold	117
4	3/25	Muffin	\$3.00	12	\$36.00		Total # of Cupcakes Sold	291
5	3/26	Cupcake	\$2.00	40	\$80.00		Total # of Muffins Sold	54
6	3/26	Pie	\$5.00	15	\$75.00		Total # of Pies Sold	66
7	3/27	Cupcake	\$2.00	35	\$70.00			
8	3/27	Cookie	\$1.00	25	\$25.00			
9	3/27	Muffin	\$3.00	14	\$42.00			
10	3/28	Cupcake	\$2.00	32	\$64.00			
11	3/28	Pie	\$5.00	16	\$80.00			
12	3/29	Cupcake	\$2.00	38	\$76.00			
13	3/29	Cookie	\$1.00	22	\$22.00			
14	3/29	Muffin	\$3.00	13	\$39.00			
15	3/30	Cupcake	\$2.00	36	\$72.00			
16	3/30	Pie	\$5.00	17	\$85.00			
17	3/31	Cupcake	\$2.00	39	\$78.00			
18	3/31	Cookie	\$1.00	24	\$24.00			
19	3/31	Muffin	\$3.00	15	\$45.00			
20	3/31	Pie	\$5.00	18	\$90.00			
21	3/31	Cupcake	\$2.00	41	\$82.00			
22	3/31	Cookie	\$1.00	26	\$26.00			
23								

7. The bakery manager would also like to know how much revenue each product brought in.

8. Start by making the data clear and easy to read. In cell **G9**, enter **Cookie Revenue**. In cell **G10**, enter **Cupcake Revenue**. In cell **G11**, enter **Muffin Revenue**. In cell **G12**, enter **Pie Revenue**.

9. To calculate the total revenue of the cookies, multiply the total number of cookies sold by the price of each cookie.

10. Select cell **H9**, enter **=H3*1**, and press **Enter**.

Bakery Sales March 2020

	A	B	C	D	E	F	G	H
1	Date	Name	Price	Quantity	Revenue			
2	3/25	Cupcake	\$2.00	30	\$60.00			
3	3/25	Cookie	\$1.00	20	\$20.00		Total # of Cookies Sold	117
4	3/25	Muffin	\$3.00	12	\$36.00		Total # of Cupcakes Sold	291
5	3/26	Cupcake	\$2.00	40	\$80.00		Total # of Muffins Sold	54
6	3/26	Pie	\$5.00	15	\$75.00		Total # of Pies Sold	66
7	3/27	Cupcake	\$2.00	35	\$70.00			
8	3/27	Cookie	\$1.00	25	\$25.00			
9	3/27	Muffin	\$3.00	14	\$42.00		Cookie Revenue	=H3*1
10	3/28	Cupcake	\$2.00	32	\$64.00		Cupcake Revenue	
11	3/28	Pie	\$5.00	16	\$80.00		Muffin Revenue	
12	3/29	Cupcake	\$2.00	38	\$76.00		Pie Revenue	
13	3/29	Cookie	\$1.00	22	\$22.00			
14	3/29	Muffin	\$3.00	13	\$39.00			
15	3/30	Cupcake	\$2.00	36	\$72.00			
16	3/30	Pie	\$5.00	17	\$85.00			
17	3/31	Cupcake	\$2.00	39	\$78.00			
18	3/31	Cookie	\$1.00	24	\$24.00			
19	3/31	Muffin	\$3.00	15	\$45.00			
20	3/31	Pie	\$5.00	18	\$90.00			
21	3/31	Cupcake	\$2.00	41	\$82.00			
22	3/31	Cookie	\$1.00	26	\$26.00			
23								

11. Finally, calculate the revenue of the remainder of the products (cupcakes, muffins, and pies) sold by using the same steps.

	A	B	C	D	E	F	G	H
L24	Date	Name	Price	Quantity	Revenue			
1								
2	3/25	Cupcake	\$2.00	30	\$60.00			
3	3/25	Cookie	\$1.00	20	\$20.00		Total # of Cookies Sold	117
4	3/25	Muffin	\$3.00	12	\$36.00		Total # of Cupcakes Sold	291
5	3/26	Cupcake	\$2.00	40	\$80.00		Total # of Muffins Sold	54
6	3/26	Pie	\$5.00	15	\$75.00		Total # of Pies Sold	66
7	3/27	Cupcake	\$2.00	35	\$70.00			
8	3/27	Cookie	\$1.00	25	\$25.00			
9	3/27	Muffin	\$3.00	14	\$42.00		Cookie Revenue	117
10	3/28	Cupcake	\$2.00	32	\$64.00		Cupcake Revenue	582
11	3/28	Pie	\$5.00	16	\$80.00		Muffin Revenue	162
12	3/29	Cupcake	\$2.00	38	\$76.00		Pie Revenue	330
13	3/29	Cookie	\$1.00	22	\$22.00			
14	3/29	Muffin	\$3.00	13	\$39.00			
15	3/30	Cupcake	\$2.00	36	\$72.00			
16	3/30	Pie	\$5.00	17	\$85.00			
17	3/31	Cupcake	\$2.00	39	\$78.00			
18	3/31	Cookie	\$1.00	24	\$24.00			
19	3/31	Muffin	\$3.00	15	\$45.00			
20	3/31	Pie	\$5.00	18	\$90.00			
21	3/31	Cupcake	\$2.00	41	\$82.00			
22	3/31	Cookie	\$1.00	26	\$26.00			

Pro Tip: Save the activity template

Be sure to save a copy of the spreadsheet template you used to complete this activity. You can use it for further practice or to help you work through your thought processes for similar tasks in a future data analyst role.

1.

Question 1

Reflection

In this activity, you used formulas to calculate the total number of products sold of each type and the total revenue of each. In the space provided below, write 2-3 sentences (40-60 words) to respond to each of the following questions:

- Which product did the bakery sell most? Would you recommend any changes to the sales strategy of the bakery based on this information?
- Which product generated the most revenue for the bakery? Based on this finding, what actions would you recommend that the bakery take in order to improve its profitability?
- Consider how you used formulas in this activity. In what ways will formulas enhance your effectiveness as a data analyst? What are the advantages of formulas for data analysis?

Answer:

Cupcakes are the product the bakery sells most among all of the products.

If the bakery reduced the price of pie and muffins, then their sales would be higher in that section.

Cupcake generated the most revenue for the bakery.

It seems that the cupcake is the most popular product of the bakery among all of the products. If the bakery increases the price of the cupcake, then the bakery will generate more profit.

Here I used the formula for generating revenue, the total number of each product sold, and the total revenue generated by each of the products.

Applying the formula, I can calculate the revenue, the total number of each product sold, and the total revenue generated by each of the products very easily in just a few seconds and get the answer very easily.

Applying the formula, I get the insight of the revenue of each bakery item, the total number of products sold, and also the total revenue generated by each of the products. By this we get a clear picture of bakery sales in March 2020.

Correct

Great job on analyzing the data and making recommendations that the bakery can leverage to improve its operations. As you continue to develop your skills as a data analyst, this experience will prove invaluable. It demonstrates your proficiency in using formulas to extract insights from data and make informed decisions. This is a skill that is highly transferable to real-world scenarios in your future career. Keep honing your analytical abilities, as they will be a key asset to your professional journey.